higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

CIVIL AND STRUCTURAL STEEL WORK DETAILING

(First Paper)

NQF LEVEL 3

15 MARCH 2019

This marking guideline consists of 4 pages.

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NOTE TO MARKERS:
This marking guideline is NOT absolute and serves only as a guideline. Consider correct alternative answers.

QUESTION 1: GENERAL

1.1 K
1.2 D
1.3 M
1.4 G
1.5 C
1.6 A
1.7 J
1.8 L
1.9 O
1.10 B
1.11 E
1.12 N
1.13 I
1.14 H
1.15 F

(15 × 1) **[15]**

QUESTION 2: REINFORCED-CONCRETE DETAILING

2.1
- It provides better estimation of the reinforcement-steel requirement.
- It provides better stock management for reinforcement.
- It is very useful during auditing of reinforcement.
- It can be used for reinforcement cutting, bending and making a skeleton.
- It provides quality control at the site as reinforcement is provided.
- It saves steel-reinforcement material.
- Cutting and bending can be done at the factory and transported to the site.
- It provides benchmarks for quantity and quality requirements for reinforcement.

(Any 3 × 2) (6)

2.2
- Hot-rolled bars
- Wire mesh
- Welded wire fabric
- Steel fibre and cold-worked deformed bars

(Any 3 × 1) (3)

2.3
- Beam reinforcement
- Floor slabs
- Columns
- Staircases
- Retaining walls
- Foundations

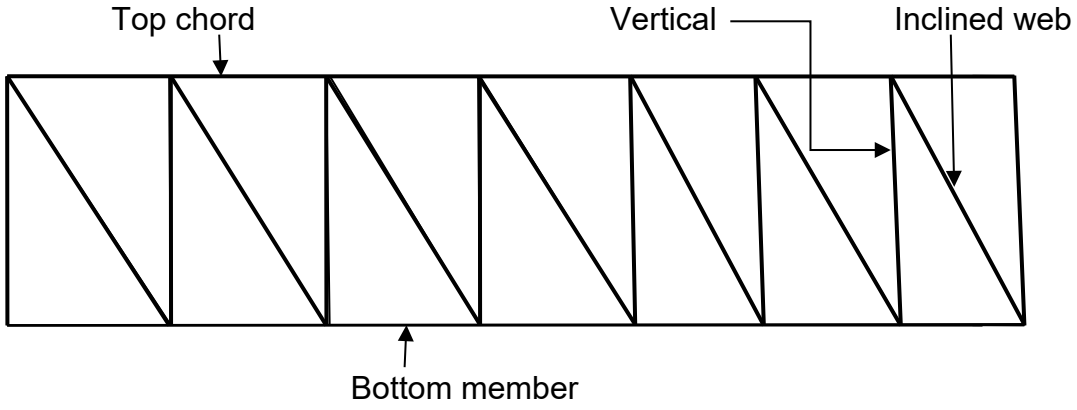
(Any 5 × 1) (5)

2.4 R is mild steel and Y is higher-yield steel. (2)
[16]

QUESTION 3: STRUCTURAL STEEL WORK DETAILING

3.1
- Corrugated galvanised steel
- Blend of zinc, steel, aluminium and silicon-coated
- Metal tile sheets
- Aluminium
- Copper
- Stone-coated steel (Any 4 × 1) (4)

3.2 N-type or Pratt truss lattice girder



Correctness	2
Neatness	1
Labelling	2 (½ mark each)

(5)

3.3
- It can enclose a large area at relatively low cost.
- Maintenance is low.
- Steel is versatile, strong and relatively cost-effective.
- No curing time is needed.
- Site assembly is easy using quickly trained semiskilled labour.
- It is quicker to fabricate and erect than building walls.
- It provides a standard range of manufacturer systems. (Any 3 × 2) (6)

3.4

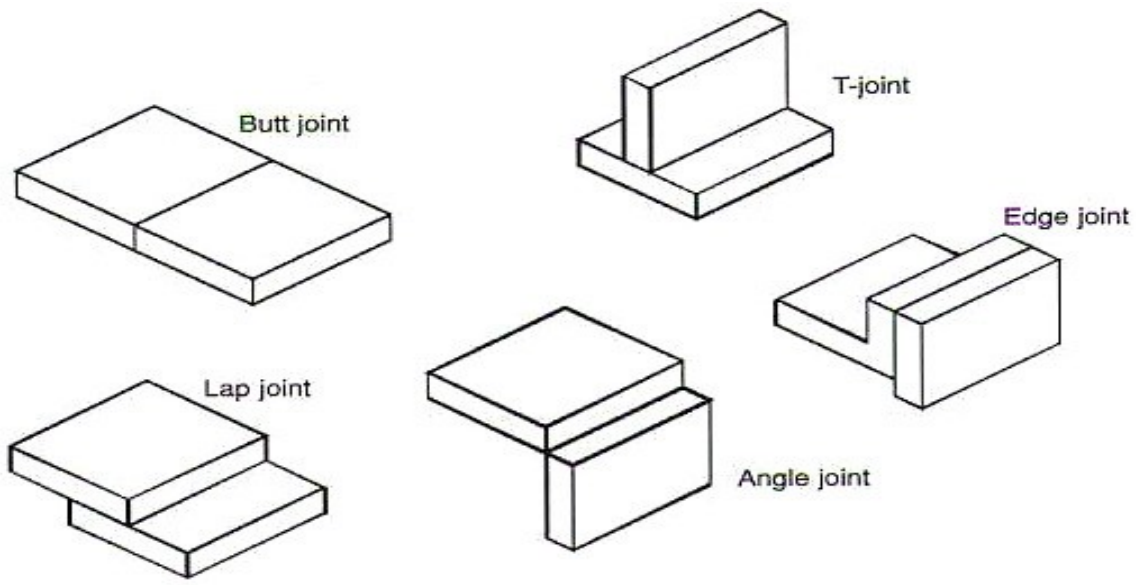
1 Member	2 Bar mark	3 Type and size	4 No. of members	5 No. in each	6 Total number	7 Length of each bar	8 Shape	2 marks per row
	1	Y30	1	6	6		⌝62	(2)
	2	Y25	1	4	4		⌊35	(2)
	3	Y20	1	3	3		⌊35	(2)
	4	Y10	1	8	8		20	(2)
	5	Y16	1	2	2		⌊35	(2)
	6	Y16	1	8	8		20	(2)
	7	Y16	1	5	5		⌝62	(2)

(Markers must use their discretion in allocating marks per row) (14)
[29]

QUESTION 4: WELDING

4.1
- Butt joint
- Lap joint
- T-joint
- Edge joint
- Angle joint (5)

4.2



(1 for drawing + 1 for labelling)
(5 × 2) (10)
[15]

TOTAL: 75

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